IAT New Test 17

Question paper, answer key and worked solutions

60 questions 180 minutes 240 marks

Biology 1–15; Chemistry 16–30; Mathematics 31–45; Physics 46–60.

Correct +4; incorrect -1; unattempted 0. All 60 questions are available.

This is a provisional 2026-pattern authoring draft for 2027 preparation. Independent academic review is pending.

Instructions

Each item has exactly one correct option. Use the separate solution section for review after attempting the paper. **Figures and equations are part of the questions.**

Biology

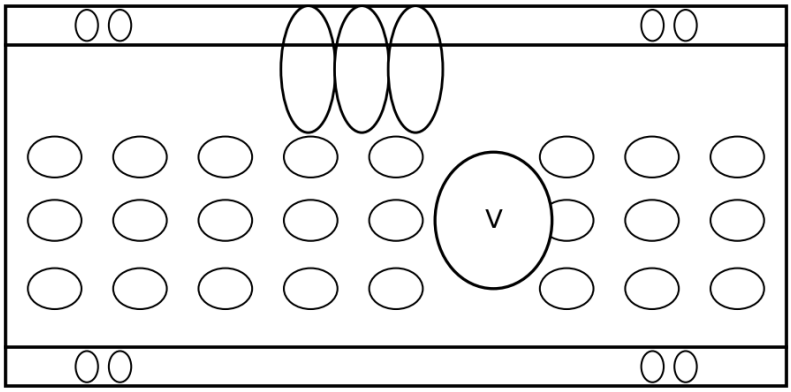
Q1 Mature pea seeds store much food in cotyledons with little residual endosperm; mature maize retains a large endosperm. Which conclusion is correct?
A) Only maize can be an angiosperm seed
B) Pea never formed endosperm
C) Maize endosperm is maternal seed-coat tissue
D) Endosperm can be consumed during development; absence at maturity does not imply failure to form

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Q2 In the leaf section, both surfaces have stomata and mesophyll is not differentiated into palisade and spongy layers. Large thin-walled cells group in the upper epidermis; their loss of turgor accompanies rolling. Which interpretation fits?

Leaf transverse section

Enlarged upper epidermal cells



V: vascular bundle; similar mesophyll throughout

A) A dorsiventral dicot leaf; the large cells are root endodermis
B) A C4 bundle sheath explains all labelled features
C) An isobilateral monocot leaf with bulliform cells
D) A stem with vascular cambium

Q3 Two nuclei each have DNA mass 4C; a normal diploid G1 nucleus has 2C. P is an ordinary diploid G2 nucleus. Q results from fusion of two diploid G1 nuclei without intervening replication. Counting chromosomes by centromeres, with normal diploid number 2n, which comparison is correct?
A) P has 2n duplicated chromosomes; Q has 4n unduplicated chromosomes
B) Both have 2n duplicated chromosomes because DNA masses match
C) P has 4n unduplicated chromosomes; Q has 2n duplicated chromosomes
D) All 4C nuclei must be tetraploid

Q4 A simplified model supplies each leaf 90 ATP and 60 NADPH per interval. The Calvin cycle requires 9 ATP and 6 NADPH per gross triose phosphate made from three CO2. A C4 pump adds 2 ATP per CO2 and eliminates the stated carbon loss. A C3 leaf loses fraction p of gross triose-equivalent carbon to photorespiration. Ignore any additional energy cost of that loss. At what p are net triose-equivalent outputs equal?
A) 1/5
B) 2/5
C) 1/2
D) 3/5

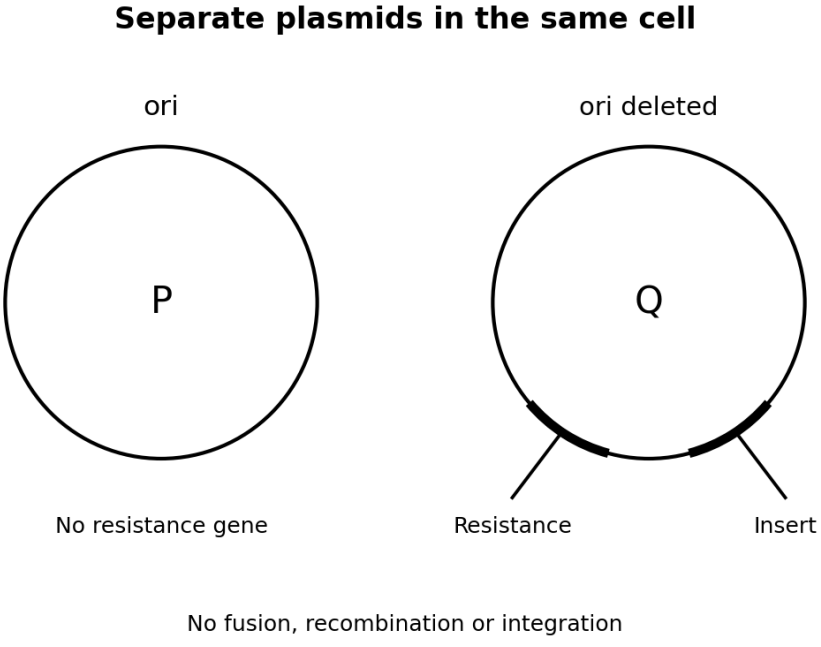
Q5 Left atrial, left ventricular and aortic pressures are respectively 8, 50 and 80 mmHg. Ventricular pressure is rising immediately after ventricular excitation. Valves respond to pressure differences. Which phase and valve state fit?
A) Ejection; mitral and aortic open
B) Filling; mitral open, aortic closed
C) Relaxation; mitral closed, aortic open
D) Isovolumetric contraction; both closed

Q6 A pollen tube enters a typical angiosperm embryo sac near the micropyle. Which cells normally guide entry using a filiform apparatus?
A) Synergids
B) Antipodals
C) Seed-coat cells
D) Endosperm cells already formed by triple fusion

Q7 A bacterium has functional lac promoter, operator, repressor and lacZ but nonfunctional lactose permease. Lactose fails to induce beta-galactosidase. An inducer entering independently of permease restores synthesis; lacY complementation also restores induction by lactose. Which inference is best supported?
A) The lacZ coding sequence is deleted
B) The operator cannot bind repressor
C) Failure of inducer access explains the phenotype better than failure of transcription machinery
D) The inducer must chemically repair permease

Q8 Aerobic flocs lower sewage BOD. Settled activated sludge is then transferred to a separate oxygen-free digester that collects gas. Which assignment is consistent?
A) Methane is produced by oxygen-dependent methanogens in the aeration tank
B) Anaerobic sludge digestion produces biogas, distinct from aerobic BOD reduction
C) Primary settling directly converts all dissolved organics to methane
D) Low-BOD effluent must be sterile

Q9 Plasmid P in the figure has ori but no resistance gene. Q has an insert and resistance gene but no ori. Both enter one cell. Assume ori acts only on linked DNA, resistance protein can protect the whole cell, and recombination, integration and fusion are absent. After many divisions and repeated dilution of old DNA/protein, which change most directly permits stable recovery of Q with its insert under selection?



A) Restore functional ori on Q
B) Add a second resistance gene to Q without ori
C) Add more P molecules only
D) Remove Q's resistance gene without restoring ori

Q10 A barnacle survives in the lower shore after a competitor is removed, but dies in the upper shore whether that competitor is present or absent. Removal does not change abiotic conditions. Which conclusion is best supported?
A) Competition excludes it from both zones
B) The two species necessarily have identical physiological limits
C) The barnacle always benefits from the competitor
D) Competition narrows its realised range, while upper-shore failure needs another explanation

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Q11 UUU and UUC both specify phenylalanine, while each codon has only one amino-acid meaning in the standard code. Which pair of properties is illustrated?
A) Overlapping and ambiguous
B) Degenerate and unambiguous
C) Universal and overlapping
D) Ambiguous and non-degenerate

Q12 A beta-globin mRNA codon changes from 5'-GAG-3' to 5'-GUG-3'. Given GAG=glutamate and GUG=valine, which description is correct?
A) Template 3'-GAG-5' becomes 3'-GUG-5'; frameshift
B) The DNA template is unchanged
C) Template 3'-CTC-5' becomes 3'-CAC-5'; one amino acid is replaced
D) Three nucleotides must be deleted

Q13 Two arterial samples have equal dissolved oxygen concentration and pO2. In Q, carbon monoxide makes half the haemoglobin oxygen-binding capacity unavailable; in P it is available. Remaining sites have equal oxygen occupancy. Which inference follows?
A) Q can have less total oxygen despite the same pO2
B) Equal pO2 guarantees equal total oxygen
C) Q must have twice as much dissolved oxygen
D) Q must carry all oxygen as bicarbonate

Q14 Cultured fibroblasts stop dividing after serum growth factors are withdrawn and retain normal G1 DNA content. Restoring serum makes many enter S phase and subsequently divide. Which interpretation is best supported?
A) All G1-DNA cells are irreversibly differentiated
B) An unnoticed meiosis must have occurred
C) Serum first doubles chromosome number by nuclear fusion
D) The arrested cells retained capacity to re-enter the proliferative cycle

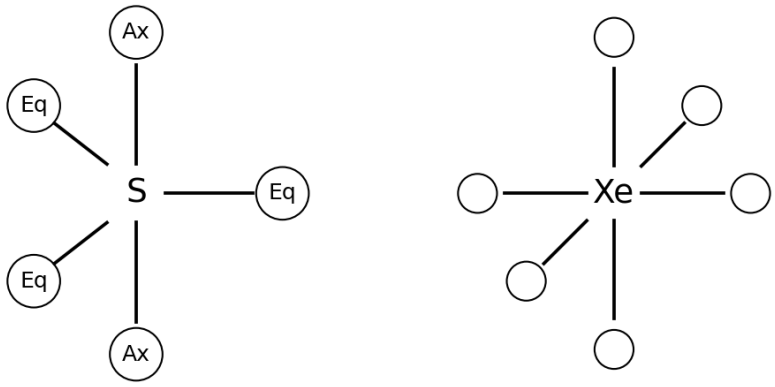
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Q15 In an illuminated chloroplast, photosystem II is blocked but a closed cyclic electron route around photosystem I still builds a proton gradient. ATP synthase works and no external electron donor is supplied. Which sustained output is compatible with that route?
A) ATP, NADPH and oxygen in the normal noncyclic ratio
B) ATP without net NADPH production or oxygen evolution
C) NADPH and oxygen but no ATP
D) Oxygen alone

Chemistry

Q16 The figure gives the electron-domain arrangements of SF4 and XeF4, with central-atom lone-pair positions left unassigned. Which placement minimises repulsion in the standard VSEPR model?

Candidate electron-domain sites



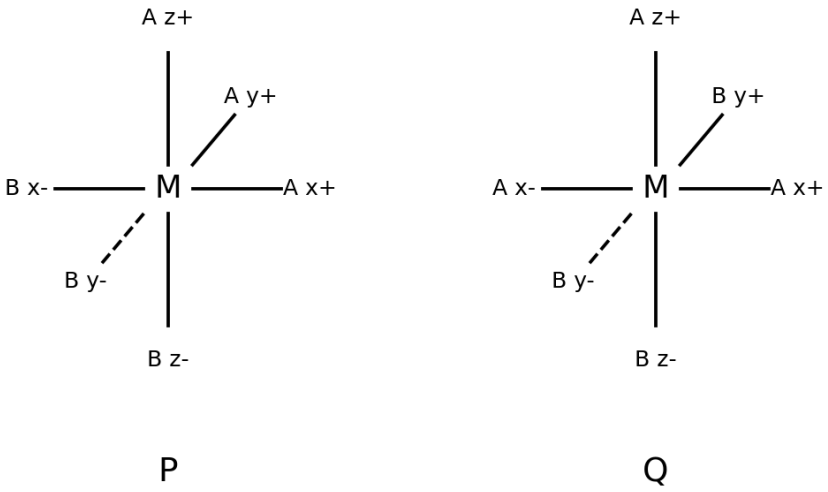
4 F + 1 lone pair

4 F + 2 lone pairs

A) SF4 axial lone pair; XeF4 adjacent lone pairs
B) SF4 equatorial lone pair; XeF4 adjacent lone pairs
C) SF4 equatorial lone pair; XeF4 opposite lone pairs
D) SF4 axial lone pair; XeF4 no lone pairs

Q17 An octahedral complex has three identical A and three identical B monodentate ligands. In arrangement P, A occupies +x,+y,+z sites. In Q, A occupies +x,-x,+z. Opposite axis sites are trans. Which classification follows?

Octahedral coordinate occupancies



A) P meridional; Q facial
B) Both facial
C) Both meridional
D) P facial; Q meridional

Q18 In acidic solution one dichromate ion is reduced completely to two Cr3+ ions. Chromium has Z=24; assume aqueous Cr3+ has three unpaired electrons and use the spin-only model. Which tuple gives (initial d count per Cr, electrons accepted per dichromate, final moment per Cr in Bohr magnetons)?
A) $(0, 3, \sqrt{15})$
B) $(0, 6, \sqrt{15})$
C) $(3, 6, \sqrt{3})$
D) $(6, 12, \sqrt{48})$

Q19 For an atomic orbital the radial-node count is n-l-1. Consider all orbitals with n=3 and exactly one radial node. How many electrons with spin quantum number +1/2 can occupy these orbitals in one atom without violating Pauli exclusion?
A) 3
B) 6
C) 9
D) 18

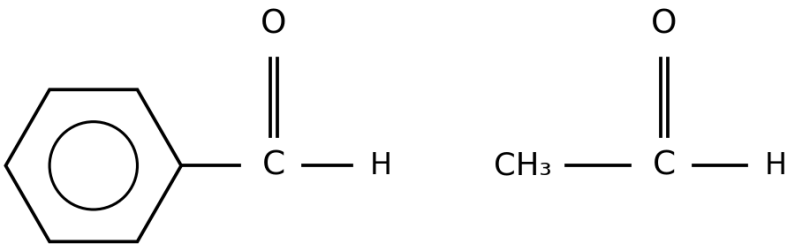
Q20 Compare the first electron-gain enthalpies of F and Cl, and the addition of a second electron to gaseous O-. Which statement fits the usual atomic trends?
A) F's first gain is more exothermic because its atom is smaller; the second gain by O- is exothermic
B) Cl's first gain is less exothermic; the second gain by O- is endothermic
C) Cl's first gain is more exothermic; the second gain by O- is endothermic
D) Cl's first gain is more exothermic; the second gain by O- is exothermic

Q21 Complete combustion of 0.900 g of a compound containing only C,H,O gives 1.320 g CO2 and 0.540 g H2O. Its molar mass is 90.0 g/mol. Use atomic masses C=12,H=1,O=16. What is its molecular formula?
A) C2H2O4
B) C3H6O3
C) CH2O
D) C3H8O2

Q22 How many constitutional isomers with formula C4H10O are acyclic ethers? Ignore conformers and stereoisomers.
A) 1
B) 2
C) 4
D) 3

Q23 The structures show benzaldehyde and ethanal. Under controlled crossed-aldol conditions, only the ethanal enolate acts as donor and benzaldehyde as acceptor. The addition product is dehydrated, then its remaining carbonyl is reduced with NaBH4 without reducing C=C. Which final product and ordinary Tollens response fit this sequence?

The controlled crossed-aldol reactants



Benzaldehyde Ethanal

A) Ph-CH=CH-CH2OH; negative
B) Ph-CH=CH-CHO; positive
C) Ph-CH2-CH2-CH2OH; negative
D) Ph-CH2-CH(OH)-CH3; negative

Q24 Sodium phenoxide is treated with CO2 under the usual Kolbe reaction conditions, then acidified. Which product relation and bicarbonate response are expected for the principal product?
A) Phenol alone; no CO2 with bicarbonate
B) An ortho-hydroxybenzoic acid; CO2 with bicarbonate
C) A para-hydroxybenzaldehyde; CO2 with bicarbonate
D) An ortho-hydroxybenzoic acid; no CO2 with bicarbonate

Q25 Ethylamine is treated with nitrous acid under standard cold deamination conditions; no aromatic ring is present. Which principal organic product and gas are expected?
A) Ethyl diazonium chloride retained indefinitely; no gas
B) Ethene only; hydrogen
C) Ethanol; nitrogen
D) Nitroethane; oxygen

Q26 A linear tripeptide contains one residue each of valine, alanine and glycine. End-group analysis identifies valine at the N terminus and glycine at the C terminus. What are its sequence from N to C and the number of water molecules required for complete peptide-bond hydrolysis per molecule?
A) Val-Ala-Gly; 2
B) Gly-Ala-Val; 2
C) Val-Gly-Ala; 3
D) Val-Ala-Gly; 3

Q27 At constant temperature a gas obeys Henry's law in a dilute solution. The gas phase changes from total pressure 5 atm and solute-gas mole fraction 0.20 to total pressure 10 atm and mole fraction 0.050. Henry's constant is unchanged. What is the final dissolved mole fraction divided by its initial value?
A) 4
B) 2
C) 1
D) 1/2

Q28 The same charge passes through an ideal Ag+ coulometer and a Cu2+ plating cell. The coulometer deposits 1.08 g Ag. The copper cell deposits 0.254 g Cu; its only other cathodic reaction is hydrogen formation from H+. Use Ag=108, Cu=63.5 g/mol, and complete current recovery. What are copper current efficiency and moles H2 evolved?
A) 40%; 0.003 mol
B) 80%; 0.001 mol
C) 80%; 0.002 mol
D) 100%; zero

Q29 A signal for a first-order disappearance reaction is $S(t) = b + ae^{-kt}$, with constant unknown background b and positive a,k. Measurements at 0,10,20 min are 12,8,6 units. Which pair gives half-life and b?
A) 20 min; 0 units
B) 10 min; 0 units
C) 10 min; 4 units
D) 5 min; 4 units

Q30 An ideal-gas equilibrium $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ is at fixed temperature in a rigid vessel. Helium is added without changing volume or reacting. Immediately after addition, which statement is correct?

A) The reaction quotient in partial pressures is unchanged, so there is no induced shift

B) Total pressure increases, so the equilibrium must shift left

C) All reacting-gas partial pressures decrease, so it shifts right

D) The equilibrium constant increases with total pressure

Mathematics

Q31 A circle passes through the origin and both intersection points of $x^2+y^2=25$ and $(x-4)^2+y^2=9$. What is its centre?
A) (4,0)
B) (25/8,0)
C) (3,0)
D) (0,25/8)

Q32 What is the maximum of $f(x)=x/(1+x^2)$ on the closed interval [0,2]?
A) 1/2
B) 2/5
C) 1
D) 0

Q33 The curve $xe^y+ye^x=1$ passes through (1,0). Where does its normal at that point meet the y-axis?
A) (0,1+e)
B) (0,-1/(1+e))
C) (0,1/(1+e))
D) (0,-(1+e))

Q34 Evaluate $\int_0^1 \frac{x^2+1}{x+1}\,dx$.
A) $2\ln 2+1/2$
B) $\ln 2-1/2$
C) $2\ln 2-1/2$
D) 1/2

Q35 Let $f(x)=\sqrt{(x-1)/(x+1)}$ with its maximal real domain and the nonnegative square root. What are the domain and range of $f\circ f$?
A) $(-\infty,-1)\cup[1,\infty);\ [0,\infty)$
B) $(-\infty,-1);\ (0,1)$
C) $(-\infty,-1];\ [0,1]$
D) $[1,\infty);\ [0,1)$

Q36 Unit vectors a,b satisfy $a\cdot b=1/2$. A vector c lies in their span and obeys $c\cdot a=1,\ c\cdot b=0$. What is |c|?
A) 1/2
B) 1
C) 2
D) $2/\sqrt{3}$

Q37 A 3x3 determinant with rows a,b,c is 5. What is the determinant with rows a+b, b+c, c+a?
A) 10
B) 5
C) 0
D) 20

Q38 A coin is chosen once: a fair coin with probability 2/3 or a double-headed coin with probability 1/3. The chosen coin is tossed independently twice, giving two heads. It is retained for a third toss. What is the probability the third toss is a tail?
A) 1/3
B) 1/4
C) 1/6
D) 1/2

Q39 How many onto functions map a four-element labelled set to a three-element labelled set?
A) 81
B) 48
C) 24
D) 36

Q40 For a positive integer n, the sum of coefficients of odd powers in (1+x)^n is 64. What is the coefficient of x^4 in $(1-x/2)^n$?
A) $-35/16$
B) $35/16$
C) $35/8$
D) $35/32$

Q41 Positive increasing a,b,c form an arithmetic progression with sum 21. The three numbers a-1,b-1,c+1 form a geometric progression. What is the arithmetic common difference?
A) $1+\sqrt{13}$
B) $\sqrt{13}$
C) $\sqrt{13}-1$
D) 3

Q42 For which real a does $x^2-2a|x|+a+2=0$ have four distinct real solutions, all strictly between -3 and 3?
A) $(2,11/5)$
B) $[2,11/5]$
C) $(-\infty,-1)\cup(2,\infty)$
D) $(2,3)$

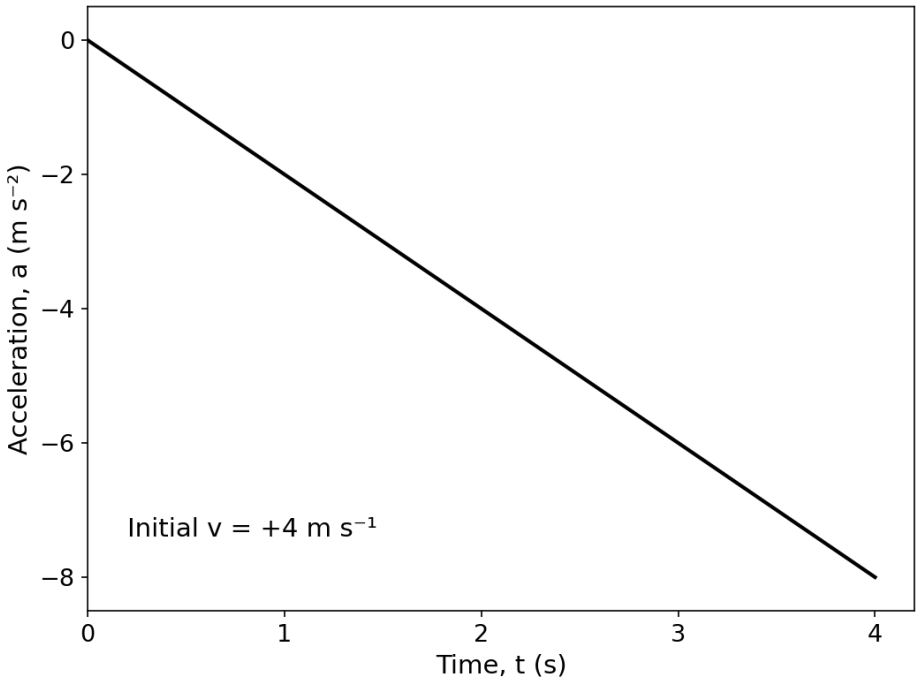
Q43 Five recorded observations have mean 4 and population variance 10. One recorded value 10 should have been 0; the other four are correct. What is the corrected population variance?
A) 10
B) 6
C) 4
D) 2

Q44 The solution of $y'-2y=e^{2x}$, y(0)=1, is considered on the real line. What is its global minimum value?
A) $-e^{-3}$
B) $-1/(2e^3)$
C) $-1/(2e)$
D) 0

Q45 Define $F(x) = x^{-1} \int_0^x \cos(t^2)\, dt$ for x not zero. What value of F(0) makes F continuous at zero?
A) 0
B) -1
C) 1
D) No such value exists

Physics

Q46 The graph shows a(t)=-2t m/s2 for 0<=t<=4 s. A particle starts at x=0 with velocity +4 m/s. What are its total distance and signed displacement over these four seconds?



A) 16 m; -16/3 m
B) 16/3 m; -16/3 m
C) 16 m; +16/3 m
D) 32/3 m; -16/3 m

Q47 A solid's mass is (10.0 +/- 0.1) g. Liquid readings before and after immersion are (50.0 +/- 0.1) and (55.0 +/- 0.1) cm3. Using maximum first-order uncertainty propagation for density, what is its percentage uncertainty?
A) 4.1%
B) 5.0%
C) 4.0%
D) 1.0%

Q48 A small sphere of density twice the liquid density falls at terminal speed vT under gravity and Stokes drag. It is given an instantaneous upward velocity of the same magnitude vT, without changing the fluid or sphere. Immediately afterward, what is its acceleration?
A) Zero
B) g/2 downward
C) g downward
D) 2g downward

Q49 A free uniform rod of mass M and length L has moment of inertia ML2/12 about its centre. Equal forces F act perpendicular to it at its two ends in opposite directions, forming a couple. What are centre-of-mass acceleration and angular-acceleration magnitude?
A) F/M; zero
B) 2F/M; 12F/(ML)
C) Zero; 6F/(ML)
D) Zero; 12F/(ML)

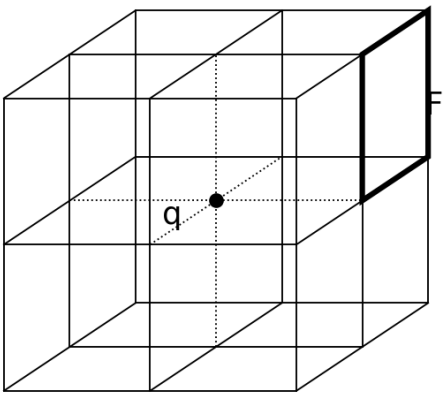
Q50 Identical ideal monatomic-gas samples start at T0,V0. P freely expands into vacuum inside an insulated rigid enclosure to 8V0. Q expands reversibly and adiabatically to 8V0, with gamma=5/3. What is final temperature TP/TQ?
A) 4
B) 1
C) 1/4
D) 8

Q51 A particle executes SHM with amplitude A and period T. At t=0 it is at x=A/2 and moving toward decreasing x. What is the first time it reaches x=-A/2?
A) T/3
B) T/2
C) T/6
D) 2T/3

Q52 A string fixed at both ends has consecutive natural frequencies 120 and 160 Hz. A point driver is then applied at its centre; assume it cannot excite a mode with a node at that point. What is the lowest natural frequency above 160 Hz that the driver can excite?
A) 180 Hz
B) 200 Hz
C) 240 Hz
D) 320 Hz

Q53 A charge q sits at the common corner of eight identical cubes assembled into one large cube, so q is at the large cube's centre. Face F in the figure is one small-cube face lying on the large cube's outer surface. What is the outward flux through F?

Eight small cubes surround the central charge

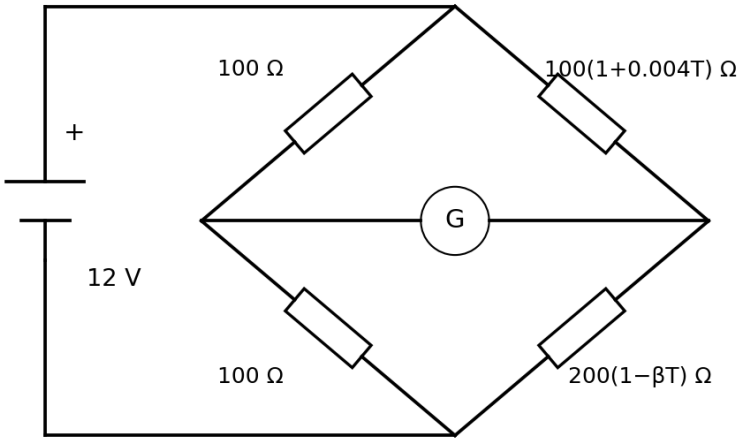


Each small cube has side a; q is the common corner

A) $q/(6\varepsilon_0)$
B) $q/(8\varepsilon_0)$
C) $q/(24\varepsilon_0)$
D) $q/(48\varepsilon_0)$

Q54 A 12 V ideal source is across a bridge’s top and bottom nodes. Left arms are both 100 ohm and temperature-independent. Right upper arm is $100(1 + 0.004T)$ ohm and right lower $200(1 - \beta T)$ ohm, T in degrees C. Only right arms share a controlled bath; this linear model holds from 0 to 50 degrees C. At T=50 the galvanometer is balanced. What are beta and total source power then?

Bridge: only the right arms share the bath



Right arms: controlled temperature T

A) 0.008 per degree C; 1.32 W
B) 0.004 per degree C; 1.44 W
C) 0.008 per degree C; 2.64 W
D) 0.016 per degree C; 1.32 W

Q55 A zero-background detector counts a mixture of A and B with half-lives 1 h and 2 h. Total activity is 900 units initially and 300 units after 2 h. Detection efficiencies are equal and constant. What fraction of total activity after 4 h comes from A?
A) 2/3
B) 1/2
C) 1/4
D) 1/3

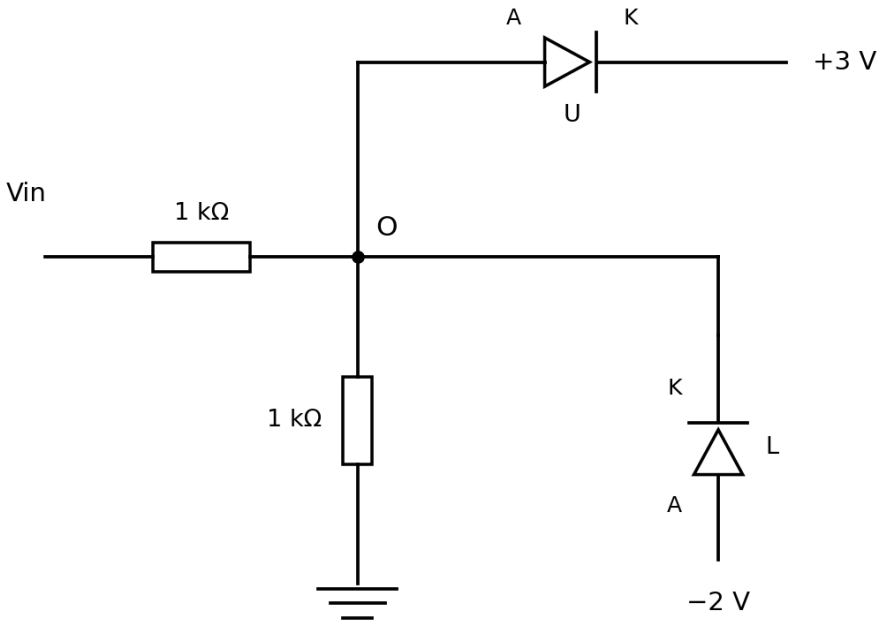
Q56 A conducting loop of area A has a small insulating gap. A uniform perpendicular magnetic field through it changes at positive rate b. Neglect capacitive transients and leakage. Which combination is correct?
A) Zero emf and zero current because the path is open
B) Emf magnitude Ab around the loop path, but no sustained conduction current across the gap
C) Emf Ab and steady current Ab/R through the insulating gap
D) Zero emf but a steady current driven by Lenz’s law

Q57 A thin converging lens of focal length +10 cm and a thin diverging lens of focal length -20 cm are in contact in air. An object is 30 cm left of them. In the thin-lens approximation, what are final image distance and signed magnification?
A) 20 cm right; -2/3
B) 60 cm right; +2
C) 60 cm right; -2
D) 30 cm right; -1

Q58 A series RLC circuit with positive R,L,C is driven at fixed rms voltage. It has equal rms currents at frequencies f and 2f. What is its resonance frequency?
A) $\sqrt{2}f$
B) $3f/2$
C) $2f$
D) $f/\sqrt{2}$

Q59 The circuit has Vin feeding node O through 1 kohm and a 1 kohm load from O to ground. Ideal diode U has anode O and cathode at a fixed +3 V reference. Ideal diode L has anode at a fixed -2 V reference and cathode O. References can source or absorb current. What are VO for Vin=+8 V and -6 V, and U current for +8 V?

Ideal diodes and a loaded output



A) +4 V, -3 V; zero
B) +3 V, -2 V; 5 mA
C) +3 V, -3 V; 2 mA
D) +3 V, -2 V; 2 mA

Q60 Two mutually coherent beams, each intensity I0 before analysis, have perpendicular linear polarisations. They overlap with variable path phase. A common ideal analyser at 45 degrees to both polarisations is placed before detection. What are maximum and minimum detected intensities as phase varies?
A) I0 and zero
B) 4I0 and zero
C) I0 and I0
D) 2I0 and zero

Answer key and worked solutions

Every solution explains the selected choice and the alternative errors.

Biology

Q1 / D Endosperm formation versus mature persistence

Answer: (D) Endosperm can be consumed during development.

Why: Angiosperm endosperm forms after triple fusion but need not persist as a large mature storage tissue. Pea largely transfers reserves to its embryo's cotyledons, whereas maize retains endosperm. This concerns tissue persistence, not whether double fertilisation occurred.

Why not the others: (A) makes retained endosperm obligatory for all angiosperms; (B) equates consumption with initial absence; (C) confuses endosperm with integument-derived seed coat.

Key point: Distinguish tissue origin from persistence at maturity.

Q2 / C Bulliform cells versus bundle sheath

Answer: (C) Isobilateral monocot leaf with bulliform cells.

Why: Both-surface stomata and undifferentiated mesophyll fit isobilateral anatomy. The large upper-epidermal cells are bulliform cells, whose turgor changes participate in rolling. They are not bundle-sheath cells around a vein, so these features alone do not establish C4 carbon fixation.

Why not the others: (A) contradicts mesophyll organisation and misplaces root endodermis; (B) confuses epidermal groups with bundle sheath; (D) ignores leaf epidermides and mesophyll.

Key point: Identify a cell's tissue position before assigning physiology.

Q3 / A Equal DNA mass from replication versus nuclear fusion

Answer: (A) P has 2n duplicated chromosomes; Q has 4n unduplicated chromosomes.

Why: Replication doubles DNA without separating sister centromeres, preserving the G2 chromosome count. Nuclear fusion instead combines two complete G1 complements, each still unduplicated. Thus equal DNA mass can arise from different chromosome numbers and ploidies.

Why not the others: (B) assumes mass fixes organisation; (C) interchanges the consequences of fusion and replication; (D) wrongly equates G2 DNA content with tetraploidy.

Key point: DNA amount, chromosome number and ploidy are distinct.

Q4 / B C3-C4 energy limitation versus photorespiratory loss

Answer: (B) p=2/5.

Given: The C4 pump adds 6 ATP per gross triose; all budgets and omitted costs are explicitly stipulated.

Formula/Principle: Take the smaller ATP-supported and NADPH-supported gross output, then apply carbon loss.

Step 1:

$$G_3 = \min(90/9, 60/6) = 10.$$

Step 2:

$$G_4 = \min(90/15, 60/6) = 6.$$

Step 3:

$$10(1-p) = 6 \Rightarrow p = 0.4 = 2/5.$$

Why: C4 avoids the stipulated loss but costs ATP, making ATP limiting. C3 starts with more gross output and ties only after losing 40%. This is a stated resource-allocation model, not a complete energy account of real photorespiration.

Why not the others: (A) leaves eight C3 units; (C) leaves five; (D) confuses retained fraction 3/5 with lost fraction.

Key point: Apply both energy limitation and carbon loss before comparing net output.

Q5 / D Valve inequalities during isovolumetric contraction

Answer: (D) Isovolumetric contraction; both valves closed.

Why: Ventricular pressure exceeds atrial pressure, closing the mitral valve, but remains below aortic pressure, preventing ejection. Rising pressure after excitation identifies contraction rather than relaxation. With both valves closed, ventricular volume is temporarily unchanged.

Why not the others: (A) requires ventricular pressure above aortic pressure and incorrectly opens mitral; (B) requires atrial pressure above ventricular; (C) contradicts the aortic inequality and rising-pressure timing.

Key point: Determine valve states from pressure inequalities before naming the phase.

Q6 / A Micropylar pollen guidance by synergids

Answer: (A) Synergids.

Why: The egg apparatus at the micropylar end contains the egg and two synergids. Synergids possess the filiform apparatus involved in guiding entry. Antipodals lie toward the chalazal end, while endosperm is a later fertilisation product rather than the pre-existing guide.

Why not the others: (B) places guidance at the wrong pole; (C) confuses maternal integument with embryo-sac cells; (D) assigns an earlier event to a later tissue.

Key point: Combine embryo-sac polarity with event order.

Q7 / C Permease-dependent access versus lac expression competence

Answer: (C) The defect is inducer access, supported by two rescue observations.

Why: Permease normally helps lactose enter. The entry bypass permits expression despite defective permease, demonstrating downstream regulatory and lacZ competence. Restoring lacY separately restores response to lactose, linking the failure to transport rather than an intrinsically unresponsive promoter.

Why not the others: (A) cannot explain restored synthesis without lacZ replacement; (B) predicts loss of repression rather than failure to induce; (D) is unnecessary because the inducer explicitly uses another entry route.

Key point: Transport bypass and complementation separate access from response.

Q8 / B Aerobic BOD removal versus anaerobic biogas

Answer: (B) Anaerobic sludge digestion produces biogas in a distinct stage.

Why: Aerobic flocs consume organic matter, reducing the effluent's oxygen demand. Settled sludge subsequently supports anaerobic digestion, including methane production. Both stages use microbes, but their oxygen conditions and material streams differ.

Why not the others: (A) gives methanogenesis the wrong oxygen condition; (C) mistakes physical settling for complete biochemical conversion; (D) confuses reduced oxygen demand with absence of organisms.

Key point: Follow the material stream and oxygen conditions separately.

Q9 / A Cis-linked replication versus transient resistance

Answer: (A) Restore ori on Q itself.

Step 1: Q can initially express resistance, but transient cell survival does not demonstrate that Q replicates.

Step 2: Under the stated linkage rule, P's origin propagates P, not an unlinked molecule in the same cytoplasm.

Step 3: Restoring ori on Q links copying capacity to Q's insert and selectable marker, allowing stable inheritance and selection.

Why: The exclusions remove integration or joining as alternative inheritance routes. Resistance acts at cell level, whereas origin function depends here on physical linkage. Repeated dilution also excludes a lasting initial protein stock as the explanation for continued selection.

Why not the others: (B) strengthens only a transient phenotype; (C) copies the wrong molecule; (D) loses selection and still fails to replicate Q.

Key point: Stable cloning needs linked replication capacity as well as selection.

Q10 / D Competitive release versus an unrescued environmental limit

Answer: (D) Competition narrows its realised range but does not explain upper-shore failure.

Why: Survival after lower-shore removal shows that zone is physiologically usable without the competitor. Its ordinary absence can therefore reflect an interaction rather than an absolute environmental barrier. Lack of upper-shore rescue shows that removing this competitor is insufficient there; another limiting factor remains.

Why not the others: (A) conflicts with failed upper-shore rescue; (B) infers the other species' limits without measurement; (C) reverses the lower-shore outcome.

Key point: An interaction is implicated where its removal changes the outcome.

Q11 / B Degeneracy versus codon ambiguity

Answer: (B) Degenerate and unambiguous.

Why: Multiple codons for one amino acid demonstrate degeneracy. A particular codon still has one assigned meaning, so the mapping is unambiguous. A many-to-one map does not imply that one input has several meanings.

Why not the others: (A) claims unsupported overlap and ambiguity; (C) infers universality across species from a codon comparison and adds overlap; (D) reverses both demonstrated properties.

Key point: Many codons for one amino acid differs from many meanings per codon.

Q12 / C Template substitution underlying a missense codon change

Answer: (C) Template CTC becomes CAC; one amino acid is replaced.

Why: The template is complementary and antiparallel to mRNA, giving the two stated DNA triplets. One base-pair substitution changes one amino-acid meaning. No base is added or removed, so downstream reading frame is preserved even though the protein sequence changes.

Why not the others: (A) copies RNA letters into DNA and invokes a length shift; (B) cannot account for the inherited codon change; (D) invents deletion when triplet length is unchanged.

Key point: Derive the antiparallel template before classifying a mutation.

Q13 / A Dissolved oxygen versus available haemoglobin capacity

Answer: (A) Q can have less total oxygen despite equal pO2.

Why: Partial pressure describes the dissolved pool, whereas much normal oxygen content is bound to haemoglobin. Removing half the available binding capacity lowers that bound component without changing the stipulated dissolved pool. Equal occupancy of surviving sites does not replace missing sites.

Why not the others: (B) equates dissolved pressure with bound-plus-dissolved content; (C) contradicts the equal dissolved concentrations; (D) assigns a major CO2 transport form to oxygen.

Key point: Oxygen pressure and total oxygen content are different measurements.

Q14 / D Reversible quiescence versus permanent differentiation

Answer: (D) The arrested cells retained proliferative capacity.

Why: DNA amount alone cannot distinguish ordinary G1 from quiescence outside the active cycle. Re-entry into S phase after growth-factor restoration is the decisive functional observation. Reversibility contradicts a blanket interpretation of non-division as permanent loss of proliferative capacity.

Why not the others: (A) contradicts observed re-entry; (B) adds reduction division without evidence; (C) confuses S-phase replication with complement fusion.

Key point: Interpret arrest using reversibility and later behaviour, not DNA content alone.

Q15 / B Cyclic electron return and ATP-only output

Answer: (B) ATP without net NADPH or oxygen.

Why: In cyclic flow, excited electrons return to photosystem I rather than ending in NADP+. Their circulation can maintain a proton gradient for ATP synthesis. Oxygen-producing water splitting belongs to photosystem II, which is blocked; the closed cycle supplies no net electrons for NADPH accumulation.

Why not the others: (A) gives noncyclic products to a cycle; (C) predicts the two absent products while omitting ATP; (D) requires the blocked water-splitting route.

Key point: Follow electron destination to distinguish ATP formation from net reducing power.

Chemistry

Q16 / C Lone-pair placement in five versus six electron domains

Answer: (C) Equatorial in SF4; opposite in XeF4.

Step 1: SF4 has four bond domains and one lone pair, giving a trigonal-bipyramidal domain arrangement. An equatorial lone pair has fewer 90-degree interactions than an axial one.

Step 2: XeF4 has four bond domains and two lone pairs in an octahedral arrangement. Opposite lone pairs maximise their separation and leave four coplanar Xe-F bonds.

Why: Electron-domain geometry includes lone pairs, whereas molecular shape describes atom positions. Minimising the strongest lone-pair repulsions gives seesaw SF4 and square-planar XeF4, rather than treating both as four-domain tetrahedral molecules.

Why not the others: (A) gives unfavourable positions in both; (B) puts the octahedral lone pairs adjacent; (D) ignores Xe's two nonbonding pairs and misplaces SF4's pair.

Key point: Count all domains, then place lone pairs before naming molecular shape.

Q17 / D Facial versus meridional occupancy from octahedral coordinates

Answer: (D) P is facial and Q meridional.

Why: In P every A-A pair occupies perpendicular axes and is cis; the three A sites bound one triangular face of the octahedron. Q has an A pair at +x and -x, which is trans, so its three A ligands lie in a meridional plane through the metal. A change of viewing direction cannot convert a trans pair to a cis pair.

Why not the others: (A) exchanges the definitions; (B) ignores Q's trans pair; (C) invents a trans A pair in P.

Key point: Use pairwise cis/trans relationships rather than the drawing's orientation.

Q18 / B Oxidation-state, electron-transfer and spin-only counts

Answer: (B) $(0, 6, \sqrt{15})$.

Given: Oxygen is -2 in dichromate; its total charge is -2. Chromium is reduced to +3.

Formula/Principle: Determine oxidation state before d count; multiply electron change by metal-atom count; use the stated spin-only relation.

Step 1:

$$2x + 7(-2) = -2 \Rightarrow x = +6.$$

Removing chromium's valence electrons to +6 leaves d0.

Step 2:

$n_e = 2(6 - 3) = 6$ electrons per dichromate.

Step 3:

$$\mu = \sqrt{n(n+2)} = \sqrt{3(5)} = \sqrt{15} \text{ BM}.$$

Why: The formula unit contains two Cr centres, but the requested final magnetic moment is per ion, not per dichromate. A coloured d0 starting species does not imply occupied d orbitals; the bookkeeping does not require assigning its colour to a d-d transition.

Why not the others: (A) forgets the second Cr in electron balance; (C) assigns the product d count to the reactant and uses one unpaired electron; (D) confuses atomic/group electron totals with oxidation-state-specific d counts.

Key point: Keep per-centre electronic structure separate from per-formula redox stoichiometry.

Q19 / A Node constraint followed by fixed-spin state counting

Answer: (A) Three electrons of the specified spin.

Given: Principal quantum number 3, radial nodes 1 and fixed spin +1/2.

Formula/Principle: Solve for l, then count allowed magnetic quantum numbers.

Step 1:

$$3 - l - 1 = 1 \Rightarrow l = 1.$$

Step 2:

$$m_l = -1, 0, +1,$$

giving three orbitals.

Step 3: Each orbital accepts only one electron with the specified spin; the second permitted electron would have the opposite spin. Thus the count is three.

Why: Specifying radial nodes selects the 3p subshell rather than the whole third shell. Fixing spin then halves the usual six-electron capacity of that subshell. Both restrictions must be applied before counting states.

Why not the others: (B) includes both spin signs; (C) counts all n=3 orbitals; (D) counts the full third-shell electron capacity.

Key point: Count distinct allowed quantum-number states after every restriction.

Q20 / C First-gain compactness exception versus second-gain repulsion

Answer: (C) Cl's first gain is more exothermic, and the second oxygen gain is endothermic.

Why: The compact fluorine 2p region produces greater electron-electron repulsion for the incoming electron than chlorine's more extended 3p region, so simple size ordering is insufficient. Adding an electron to an already negative gaseous ion also requires overcoming electrostatic repulsion. The favourable formation of an ionic solid is a different overall energy balance from this isolated gas-phase second electron gain.

Why not the others: (A) overlooks both repulsion effects; (B) reverses the F-Cl exception; (D) assigns the stabilisation of an overall compound to an isolated negatively charged ion.

Key point: Specify initial charge and electronic compactness before applying an electron-affinity trend.

Q21 / B Combustion mass balance followed by molecular multiplier

Answer: (B) C3H6O3.

Given: Oxygen belongs to the compound only after subtracting carbon and hydrogen masses; combustion oxygen is not directly a sample-oxygen measurement.

Formula/Principle: Derive elemental mole ratios from combustion products and sample mass, then use molar mass.

Step 1:

$$n_C = 1.320/44 = 0.0300, \quad n_H = 2(0.540/18) = 0.0600 \text{ mol}.$$

Step 2:

$$m_O = 0.900 - 12(0.0300) - 1(0.0600) = 0.480 \text{ g}, \quad n_O = 0.0300.$$

Step 3: The ratio is 1:2:1, so empirical mass is 30.

$$90/30 = 3 \Rightarrow \mathrm{C_3H_6O_3}.$$

Why: Product water contains two hydrogen atoms and product CO2 contains one carbon. Oxygen must be recovered by mass difference because the oxidant supplies additional oxygen. The empirical ratio alone does not determine the molecular formula.

Why not the others: (A) fits the molar mass but not measured C/H; (C) stops at the empirical formula; (D) has the wrong elemental ratio and molar mass.

Key point: Separate elemental recovery, empirical ratio and molecular multiplier.

Q22 / D Constitutional ether enumeration by unordered carbon partitions

Answer: (D) Three.

Given: An ether has a C-O-C connection and four carbons in two nonempty alkyl groups.

Formula/Principle: Partition carbons between the two oxygen-linked groups, avoiding double counting when they are interchanged.

Step 1: A 1+3 split allows methyl-n-propyl ether and methyl-isopropyl ether, two distinct connectivities.

Step 2: A 2+2 split allows diethyl ether, one connectivity. A 3+1 split repeats the first case.

Step 3:

$$N = 2 + 1 = 3.$$

Why: Branching matters for the three-carbon side but exchanging the names of the two sides does not create a new molecule. Alcohol isomers have the same formula but a different functional group and are excluded. No ring is compatible with the stated acyclic saturated ether count.

Why not the others: (A) keeps only the equal split; (B) misses the branched three-carbon group; (C) double-counts an interchange of the ether sides.

Key point: Count carbon partitions first, then distinct structures within each partition.

Q23 / A Crossed-aldol carbon tracking with selective final reduction

Answer: (A) Ph-CH=CH-CH2OH; Tollens negative. Ph denotes phenyl.

Step 1: Benzaldehyde lacks an alpha hydrogen, while ethanal provides the stipulated enolate. Its alpha carbon attacks benzaldehyde's carbonyl carbon, giving Ph-CH(OH)-CH2-CHO.

Step 2: Dehydration across the adjacent carbons gives Ph-CH=CH-CHO, preserving the ethanal-derived aldehyde.

Step 3: Selective NaBH4 reduction changes CHO to CH2OH while leaving the stated alkene intact. The final ordinary alcohol does not give the aldehyde Tollens test.

Why: Following the donor and acceptor carbons prevents attaching the new carbon-carbon bond to the wrong carbonyl. The sequence explicitly separates dehydration from carbonyl reduction and does not allow alkene hydrogenation. Each stage must be applied to the previous product.

Why not the others: (B) stops before reduction; (C) also reduces the alkene contrary to the condition; (D) moves the alcohol to a secondary carbon and removes the alkene instead of reducing only the aldehyde.

Key point: Track carbon origins through aldol addition, dehydration and selective reduction.

Q24 / B Phenoxide carboxylation followed by product acidity

Answer: (B) Ortho-hydroxybenzoic acid, giving CO2 with bicarbonate.

Why: Carboxylation of phenoxide followed by acidification gives mainly salicylic acid under the stated standard conditions. Its carboxylic-acid group can transfer a proton to bicarbonate and release CO2. This response is not the ordinary behaviour of phenol alone, so the new functional group matters as well as ring position.

Why not the others: (A) omits carboxylation; (C) confuses carboxylation with aldehyde introduction; (D) ignores the stronger carboxylic-acid proton of the product.

Key point: Reassess acid-base tests after a functional-group transformation.

Q25 / C Aliphatic primary-amine deamination rather than stable aryl diazotisation

Answer: (C) Ethanol and nitrogen.

Why: A primary aliphatic amine does not form the relatively persistent cold diazonium salt characteristic of a primary aromatic amine. Its unstable intermediate loses nitrogen and yields the alcohol in the usual aqueous reaction account. The absence of an aromatic ring is therefore important when interpreting a cold nitrous-acid test.

Why not the others: (A) transfers aryl-diazonium stability to an alkyl system; (B) predicts the wrong characteristic gas and excludes the aqueous substitution product; (D) mistakes nitrosative deamination for installation of a nitro group.

Key point: Cold conditions do not make aliphatic and aromatic diazonium behaviour identical.

Q26 / A Terminally constrained peptide reconstruction and hydrolysis

Answer: (A) Val-Ala-Gly; two water molecules.

Given: There are three distinct residues, fixed terminal identities and no cyclic bond.

Formula/Principle: Arrange residues consistently with end groups; a linear chain of n residues contains n-1 peptide links.

Step 1: Fixed N-terminal Val and C-terminal Gly leave Ala as the middle residue.

Step 2:

$$N_{\text{peptide}} = 3 - 1 = 2, \qquad N_{\text{H}_2\text{O}} = 2.$$

Why: End-group information supplies direction, not merely composition. Hydrolysis consumes one water per peptide bond, not one per free amino acid ultimately recovered. The excluded cyclic case

would have a different bond count.

Why not the others: (B) reverses the termini; (C) misplaces the C terminus and counts residues instead of bonds; (D) has the right order but the same bond-count error.

Key point: Combine residue composition with terminal direction before counting links.

Q27 / D Henry-law partial pressure under changing gas composition

Answer: (D) 1/2.

Given: The relevant gas partial pressure is mole fraction times total pressure.

Formula/Principle: In the same Henry-law convention, dissolved mole fraction is proportional to solute-gas partial pressure at fixed temperature.

Step 1:

$$p_1 = 5(0.20) = 1.0 \text{ atm}, \quad p_2 = 10(0.050) = 0.50 \text{ atm}.$$

Step 2:

$$x_2/x_1 = p_2/p_1 = 0.50.$$

Why: Doubling total pressure does not guarantee greater solubility if the gas mixture's composition also changes. Here the solute gas becomes four times more dilute in the gas phase, outweighing the pressure increase. The unchanged Henry constant cancels only after using the correct partial pressures.

Why not the others: (A) reverses the composition change; (B) uses total pressure alone; (C) assumes the pressure and composition changes cancel exactly.

Key point: Use the solute's partial pressure, not total pressure, in gas solubility.

Q28 / B Series coulometry with competing cathodic current

Answer: (B) 80%; 0.001 mol hydrogen.

Given: Silver requires one electron per atom; copper two; hydrogen two per molecule.

Formula/Principle: The ideal series coulometer measures total moles of electrons, which are then partitioned in the plating cell.

Step 1:

$$n_{e,\text{total}} = 1.08/108 = 0.0100 \text{ mol}.$$

Step 2:

$$n_{\text{Cu}} = 0.254/63.5 = 0.00400, \quad n_{e,\text{Cu}} = 0.00800 \text{ mol}.$$

Step 3:

$$\eta = 0.00800/0.0100 = 0.80, \quad n_{\text{H}_2} = (0.0100 - 0.00800)/2 = 0.00100 \text{ mol}.$$

Why: Equal series charge does not imply equal deposited atom counts or that all current deposits copper. The measured copper mass uses only part of the electron budget; the stipulated competing reaction accounts for the remainder. Dividing the residual by two converts hydrogen atoms' electron equivalents to H2 molecules.

Why not the others: (A) misses copper's two-electron stoichiometry; (C) reports residual electron moles as H2; (D) ignores the mass-based charge deficit.

Key point: Use a coulometer to establish charge before separating competing cathodic processes.

Q29 / C First-order half-life inferred with an unknown additive baseline

Answer: (C) Half-life 10 min; background 4 units.

Given: Equal time intervals share decay factor q=exp(-10k).

Formula/Principle: Differences eliminate an additive background.

Step 1:

$$S_0 - S_{10} = a(1-q) = 4, \quad S_{10} - S_{20} = aq(1-q) = 2.$$

Step 2:

$$q = 2/4 = 1/2, \quad a = 4/(1-1/2) = 8.$$

Step 3:

$$b = 12 - 8 = 4, \qquad e^{-10k} = 1/2 \Rightarrow t_{1/2} = 10 \text{ min}.$$

Why: The raw signal does not halve because it contains a constant offset. Equal-interval differences isolate the decaying component without needing b in advance. The inferred values reproduce all three observations, including the nonzero long-time signal.

Why not the others: (A) treats raw signal 12 to 6 as a concentration half-life; (B) infers the right decay interval but discards the offset; (D) doubles the decay constant incorrectly.

Key point: Remove measurement background before applying a concentration decay law.

Q30 / A Inert-gas addition at fixed volume versus compression

Answer: (A) Reaction quotient unchanged; no induced shift.

Why: At fixed temperature and volume, each reacting gas retains its mole number immediately after helium addition and hence its own partial pressure nRT/V. Total pressure increases because helium contributes another partial pressure, not because the reacting species are compressed. The reaction quotient therefore remains at the same K.

Why not the others: (B) applies compression reasoning without a volume change; (C) describes dilution by expansion rather than rigid-volume addition; (D) changes a temperature-dependent equilibrium constant without changing temperature.

Key point: Identify what happens to reacting-species partial pressures before invoking pressure effects.

Mathematics

Q31 / B Circle through a common chord and an additional point

Answer: (B) (25/8,0).

Given: Subtract the equations of the two supplied circles to locate their common chord.

Formula/Principle: Equal-distance geometry determines the centre of a circle through three non-collinear points.

Step 1:

$$(x-4)^2 - x^2 = 9 - 25 \Rightarrow -8x + 16 = -16 \Rightarrow x = 4.$$

Step 2: Then y squared=25-16=9, so the points are (4,3) and (4,-3). Their perpendicular bisector is y=0; write the required centre as (h,0).

Step 3: Equal distances to (0,0) and (4,3) give

$$h^2 = (4-h)^2 + 9 \Rightarrow 8h = 25 \Rightarrow h = 25/8.$$

Worked derivation:

$$O' = (25/8, 0), \quad R = 25/8.$$

Why: The required circle is neither supplied circle, since it must also pass through the origin. Reflection symmetry of the common points constrains the centre to the x-axis, leaving one equal-distance equation. The common chord's abscissa is not itself the new centre.

Why not the others: (A) reports the chord location; (C) mistakes the chord's half-length for the centre coordinate; (D) chooses the wrong perpendicular-bisector axis.

Key point: Use a common chord to obtain points, then impose the new circle's extra condition.

Q32 / A Rational maximum with an interior critical point and endpoints

Answer: (A) 1/2.

Given: The denominator is positive throughout the interval.

Formula/Principle: Compare stationary values with both endpoints on a closed interval.

Step 1:

$$f'(x) = \frac{1-x^2}{(1+x^2)^2} = 0 \Rightarrow x = 1 \quad \text{in } [0,2].$$

Step 2:

$$f(0) = 0, \quad f(1) = 1/2, \quad f(2) = 2/5.$$

Worked derivation:

$$\max_{[0,2]} f = 1/2.$$

Why: The function first increases and then decreases, so the largest x is not where its maximum occurs. The denominator's growth matters as much as the numerator's growth. Evaluating endpoints ensures the stationary-point calculation gives the global closed-interval result.

Why not the others: (B) checks only the right endpoint; (C) reports the maximising input rather than output; (D) reports the minimum.

Key point: Distinguish the extremising input from the extremal value.

Q33 / D Implicit exponential tangent followed by normal intercept

Answer: (D) (0,-(1+e)).

Given: At the point, exp(y)=1 and exp(x)=e.

Formula/Principle: Differentiate both products implicitly; normal slope is the negative reciprocal of nonzero tangent slope.

Step 1:

$$e^y + xe^y y' + y'e^x + ye^x = 0.$$

Step 2:

$$1 + (1+e)y' = 0 \Rightarrow m_{\text{tangent}} = -1/(1+e), \quad m_{\text{normal}} = 1+e.$$

Step 3:

$$y = (1+e)(x-1) \Rightarrow y(0) = -(1+e).$$

Worked derivation:

$$N: \; y = (1+e)x - (1+e).$$

Why: Each term depends on x both explicitly and through y, requiring the product and chain rules. The tangent's slope cannot be used for the normal. The normal's positive slope still gives a negative y-intercept because it passes through (1,0).

Why not the others: (A) drops the negative intercept sign; (B) reports tangent slope as an intercept; (C) combines reciprocal and sign errors.

Key point: Separate implicit differentiation, perpendicular slope and line intercept.

Q34 / C Polynomial division before a definite logarithmic integral

Answer: (C) 2ln2-1/2.

Given: The denominator is nonzero on [0,1].

Formula/Principle: Divide the numerator by the denominator before integrating the remainder.

Step 1:

$$\frac{x^2+1}{x+1} = x - 1 + \frac{2}{x+1}.$$

Step 2:

$$I = \left[\frac{x^2}{2} - x + 2\ln(x+1)\right]_0^1.$$

Step 3:

$$I = 1/2 - 1 + 2\ln 2 = 2\ln 2 - 1/2.$$

Worked derivation:

$$(x+1)(x-1) + 2 = x^2 + 1.$$

Why: The numerator's degree is larger, so a simple logarithm alone cannot integrate the whole expression. Polynomial division isolates the rational remainder and makes its logarithmic coefficient explicit. Checking the division also guards against losing the extra constant.

Why not the others: (A) reverses the polynomial contribution's sign; (B) loses the remainder coefficient two; (D) ignores the rational remainder.

Key point: Reduce an improper rational expression before choosing an integration technique.

Q35 / B Domain and attained range of a square-root iterate

Answer: (B) Domain (-infinity,-1), range (0,1).

Given: The intermediate value u=f(x) is nonnegative.

Formula/Principle: A composition requires the inner input and output to satisfy their respective function domains.

Step 1:

$$D_f = (-\infty, -1) \cup [1, \infty).$$

Step 2: Since u>=0, the outer domain demands u>=1.

$$\frac{x-1}{x+1} \geq 1 \iff \frac{-2}{x+1} \geq 0 \iff x < -1.$$

Step 3: On that domain u ranges over (1,infinity), with neither limiting endpoint included. For such u,

$$0 < \frac{u-1}{u+1} < 1.$$

Step 4: The outer square root continuously covers (0,1); zero would need u=1 and one would need an infinite u, neither attained.

Worked derivation:

$$D_{f\circ f} = (-\infty, -1), \qquad R_{f\circ f} = (0, 1).$$

Why: The entire domain of f is not automatically the domain of its iterate. Its nonnegative output excludes the negative outer-domain branch, and x>=1 produces an inner output below one, invalid for the second application. Tracking attained endpoints is essential for both sets.

Why not the others: (A) ignores the intermediate-domain restriction; (C) includes the undefined x=-1 and unattained range endpoints; (D) keeps precisely the inner branch that fails the outer restriction.

Key point: Propagate domains and attained ranges through each layer of a composition.

Q36 / D Reconstructing a vector from oblique dot-product constraints

Answer: (D) 2/sqrt3.

Given: Write c=alpha a+beta b; a and b are independent because their angle is 60 degrees.

Formula/Principle: Dot constraints determine coefficients in an oblique basis; their coefficients are not orthogonal components.

Step 1:

$$\alpha + \beta/2 = 1, \quad \alpha/2 + \beta = 0.$$

Step 2: Solving gives alpha=4/3 and beta=-2/3.

Step 3:

$$|c|^2 = c \cdot (\alpha a + \beta b) = \alpha(c \cdot a) + \beta(c \cdot b) = 4/3.$$

Worked derivation:

$$c = \tfrac{4}{3}a - \tfrac{2}{3}b, \qquad |c| = 2/\sqrt{3}.$$

Why: Taking alpha=1 and beta=0 would be valid for orthogonal a,b, not the stated pair. The compact norm evaluation uses the given dot products after solving, avoiding an incorrect sum of coefficient squares. The span condition excludes any unmeasured perpendicular component.

Why not the others: (A) uses the basis overlap as the vector length; (B) assumes an orthogonal basis; (C) loses the square root or normalisation in the coupled equations.

Key point: Basis coefficients and scalar projections differ for nonorthogonal directions.

Q37 / A Determinant of a simultaneous row-mixing transformation

Answer: (A) 10.

Given: The new rows are simultaneous combinations, not sequential replacements using already changed rows.

Formula/Principle: If B=MA then det B=det M det A.

Step 1:

$$M = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}, \quad \det M = 1 + 1 = 2.$$

Step 2:

$$\det B = 2(5) = 10.$$

Worked derivation:

$$\det \begin{pmatrix} a+b \\ b+c \\ c+a \end{pmatrix} = 2 \det \begin{pmatrix} a \\ b \\ c \end{pmatrix}.$$

Why: Adding a multiple of one unchanged row to another preserves a determinant, but doing all three stated combinations at once is a different linear transformation. Its determinant must be computed rather than assumed one. The original nonzero determinant also confirms the rows are independent.

Why not the others: (B) applies the single-row invariance rule indiscriminately; (C) incorrectly assumes the three sums are dependent; (D) doubles the transformation factor twice.

Key point: Distinguish simultaneous row combinations from one elementary row addition.

Q38 / C Posterior predictive probability after retaining a latent coin

Answer: (C) 1/6.

Given: The same selected coin is used throughout, and tosses are independent conditional on its identity.

Formula/Principle: Update the coin identity from the observed sequence, then average the next-toss probabilities using that posterior.

Step 1:

$$P(HH) = (2/3)(1/4) + (1/3)(1) = 1/2.$$

Step 2:

$$P(F \mid HH) = \frac{(2/3)(1/4)}{1/2} = 1/3.$$

Step 3:

$$P(T_3 \mid HH) = (1/3)(1/2) + (2/3)(0) = 1/6.$$

Worked derivation:

$$P(T_3 \mid HH) = \frac{(2/3)(1/8)}{1/2} = 1/6.$$

Why: Two heads favour the double-headed coin but do not identify it with certainty. The predictive probability needs another factor of one half after computing the posterior fair-coin probability. Selecting a fresh coin before the third toss would be a different experiment.

Why not the others: (A) stops at the posterior fair-coin probability; (B) uses a single-head update; (D) ignores the identity information.

Key point: Distinguish posterior model probability from the next-event probability.

Q39 / D Onto function count with an independent occupancy verification

Answer: (D) 36.

Given: Every one of the three target labels must appear at least once.

Formula/Principle: Count all functions, subtract those missing at least one target, and correct overlapping exclusions.

Step 1: There are 3^4=81 unrestricted functions. For each omitted target, 2^4=16 functions remain.

Step 2: Omitting two specified targets leaves one function. There are three pairs of omitted targets.

Step 3:

$$N = 3^4 - \binom{3}{1}2^4 + \binom{3}{2}1^4 = 81 - 48 + 3 = 36.$$

Worked derivation:

$$N = \binom{4}{2}\, 3\, 2! = 36,$$

equivalently choosing the unique doubled target's two preimages and assigning the other two.

Why: Subtracting three missing-target counts removes constant functions twice too often; the pairwise correction restores them appropriately. The targets are labelled, so permutations of their identities matter. The alternate double-preimage argument independently verifies the count.

Why not the others: (A) allows missing targets; (B) reports excluded-function counts rather than the answer; (C) treats onto maps like an injective selection and miscounts four inputs.

Key point: Labelled onto mappings require every target plus overlap correction.

Q40 / B Parity coefficient sum followed by scaled signed-term extraction

Answer: (B) 35/16.

Given: n>0, so evaluations at +1 and -1 separate even and odd coefficient sums.

Formula/Principle: Binomial parity sums determine n; the general term then gives the requested coefficient.

Step 1:

$$S_{\text{odd}} = \frac{(1+1)^n - (1-1)^n}{2} = 2^{n-1} = 64 \Rightarrow n = 7.$$

Step 2:

$$[x^4](1 - x/2)^7 = \binom{7}{4}(-1/2)^4 = 35/16.$$

Worked derivation:

$$n = 7, \quad (-1)^4 = 1, \quad 2^4 = 16.$$

Why: The odd sum is half the total coefficient sum because n is positive. After identifying the exponent, the selected term has an even power of the negative factor and is therefore positive. The coefficient is not the numerical value of the whole polynomial.

Why not the others: (A) keeps a negative sign for an even power; (C) divides by 2^3; (D) divides by 2^5.

Key point: Infer polynomial degree from evaluation identities before extracting a scaled term.

Q41 / C AP reconstruction followed by a perturbed three-term GP

Answer: (C) sqrt13-1.

Given: The AP middle term is 7; write a=7-d,b=7,c=7+d with d>0 and a>0.

Formula/Principle: The square of the middle GP term equals the product of the outside terms.

Step 1: The transformed terms are 6-d,6,8+d.

Step 2:

$$36 = (6-d)(8+d) = 48 - 2d - d^2 \Rightarrow d^2 + 2d - 12 = 0.$$

Step 3:

$$d = -1 \pm \sqrt{13}.$$

Only -1+sqrt13 is positive; it is below 7, so the original first term is positive.

Worked derivation:

$$d = \sqrt{13} - 1, \quad a = 8 - \sqrt{13} > 0.$$

Why: The perturbations do not preserve the original AP, so a GP condition must be imposed on the changed numbers. Both quadratic roots must be considered, then the stated order and positivity select the feasible one. This uses only finite three-term progression relations.

Why not the others: (A) reverses the linear coefficient's root sign; (B) loses the shift by one; (D) does not satisfy the exact GP equation.

Key point: Apply each progression relation to the correct, possibly transformed, terms.

Q42 / A Four-root absolute-value quadratic with strict interval constraints

Answer: (A) 2<a<11/5.

Given: Put u=|x|>=0. Four distinct x solutions require two distinct positive u roots, both below 3.

Formula/Principle: Combine discriminant, root signs, vertex position and endpoint polynomial sign.

Step 1:

$$q(u) = u^2 - 2au + a + 2, \quad \Delta/4 = a^2 - a - 2 = (a-2)(a+1).$$

Step 2: Distinct real positive roots require Delta>0, sum 2a>0 and product a+2>0. Together these give a>2.

Step 3: Both roots below 3 require the vertex a<3 and

$$q(3) = 11 - 5a > 0 \Rightarrow a < 11/5.$$

Step 4: These conditions are sufficient: the two positive distinct roots lie left of 3, each producing a positive and negative x. At a=2 roots coincide; at a=11/5 a root equals 3.

Worked derivation:

$$2 < a < 11/5, \qquad x = \pm u_1, \pm u_2.$$

Why: A discriminant test alone counts u roots, not valid absolute-value inputs or the resulting four distinct x values. Positivity avoids losing a branch at u=0. Strict location and distinctness exclude different parameter endpoints for different reasons.

Why not the others: (B) includes both invalid endpoints; (C) uses only discriminant regions; (D) enforces a vertex bound without the stricter endpoint sign.

Key point: Translate an absolute-value equation into nonnegative roots before counting and locating solutions.

Q43 / D Variance correction through both raw moments

Answer: (D) 2.

Given: Population variance divides by five, not four.

Formula/Principle: Update the raw sum and sum of squares before recomputing the changed mean and variance.

Step 1:

$$\sum x_i = 5(4) = 20, \quad \sum x_i^2 = 5(10 + 4^2) = 130.$$

Step 2: Correcting 10 to 0 gives

$$\sum x_i = 10, \quad \bar{x} = 2, \quad \sum x_i^2 = 30.$$

Step 3:

$$\sigma^2_{\text{new}} = 30/5 - 2^2 = 6 - 4 = 2.$$

Worked derivation:

$$\sigma^2_{\text{new}} = \frac{130 - 100}{5} - \left(\frac{20 - 10}{5}\right)^2 = 2.$$

Why: The changed observation alters both raw moments and the reference mean used for all deviations. Subtracting only the erroneous squared value while retaining the old mean would give an invalid result. The stated data are feasible, for example original values 10,1,2,3,4.

Why not the others: (A) assumes variance is unchanged by correcting one value; (B) stops at the new second moment; (C) reports the square of the corrected mean.

Key point: A data correction changes both first and second moments.

Q44 / B Linear initial-value solution followed by global optimisation

Answer: (B) -1/(2e^3).

Given: This is a first-order linear equation with constant coefficient -2.

Formula/Principle: Use an integrating factor to solve, then determine derivative signs for a global extremum.

Step 1:

$$\frac{d}{dx}(ye^{-2x}) = 1 \Rightarrow ye^{-2x} = x + C.$$

The initial condition gives C=1.

Step 2:

$$y = (x+1)e^{2x}, \qquad y' = e^{2x}(2x+3).$$

Step 3: Since e^(2x)>0, y decreases for x<-3/2 and increases for x>-3/2. Thus the sole critical point is the global minimum.

Step 4:

$$y(-3/2) = (-1/2)e^{-3} = -1/(2e^3).$$

Worked derivation:

$$\min_{\mathbb{R}} y = -1/(2e^3), \quad x_{\min} = -3/2.$$

Why: Solving the ODE is necessary before optimising its particular solution. Monotonicity on both sides establishes a global minimum without assuming that every stationary point is one. The negative value is consistent with the solution crossing zero at x=-1.

Why not the others: (A) omits the factor one half; (C) uses the wrong exponential argument at the critical point; (D) reports the zero crossing, not the minimum.

Key point: Carry an initial-value solution into a separate derivative-sign check.

Q45 / C Removable singularity through the fundamental theorem

Answer: (C) 1.

Given: The integrand cos(t squared) is continuous near zero; an explicit elementary antiderivative is not required.

Formula/Principle: The fundamental theorem gives the derivative of the integral at its lower-limit point.

Step 1: Let $G(x) = \int_0^x \cos(t^2)\,dt$. Then G(0)=0 and G'(0)=cos0=1.

Step 2:

$$\lim_{x\to 0} \frac{G(x)}{x} = \lim_{x\to 0} \frac{G(x) - G(0)}{x - 0} = G'(0) = 1.$$

Worked derivation:

$$F(0) = 1$$

gives the same limit from either sign of x, because the oriented integral changes sign with its interval.

Why: The expression is the average integrand value over a shrinking interval. Continuity of that integrand makes the average approach its value at zero. Failure to write an elementary antiderivative is not a failure of the definite integral or its limit.

Why not the others: (A) notices the vanishing numerator but ignores the denominator; (B) mishandles orientation for negative x; (D) confuses non-elementary antiderivative form with nonexistent limit.

Key point: A derivative-of-integral limit can avoid evaluating the integral itself.

Physics

Q46 / A Velocity reversal inferred from a linear acceleration graph

Answer: (A) Distance 16 m; displacement -16/3 m.

Given: Acceleration is a straight decreasing line, not a constant on the interval.

Formula/Principle: Integrate acceleration to locate velocity reversal, then integrate speed on the resulting sign intervals.

Step 1:

$$v(t) = 4 + \int_0^t -2s\,ds = 4 - t^2.$$

The reversal occurs at t=2 s.

Step 2:

$$x(t) = \int_0^t (4 - s^2)\,ds = 4t - t^3/3, \quad x(2) = 16/3, \quad x(4) = -16/3 \text{ m}.$$

Step 3:

$$D = |x(2) - x(0)| + |x(4) - x(2)| = 16/3 + 32/3 = 16 \text{ m}.$$

Worked derivation:

$$\Delta x = -16/3 \text{ m}, \quad D = 16 \text{ m}.$$

Why: Negative acceleration does not imply negative velocity from the start. The particle first travels forward, stops, then travels backward past its origin. Distance requires the actual turn point rather than the absolute value of final displacement.

Why not the others: (B) equates distance to displacement magnitude; (C) loses the final direction; (D) counts only the backward leg.

Key point: Find velocity zeroes before converting integrated motion to total distance.

Q47 / B Density uncertainty amplified by subtracting volume readings

Answer: (B) 5.0%.

Given: Displacement volume is the difference of two independent readings; use worst-case absolute bounds, not statistical quadrature.

Formula/Principle: Absolute uncertainty adds for a difference, and fractional uncertainty adds for a quotient.

Step 1:

$$V = 55.0 - 50.0 = 5.0 \text{ cm}^3, \quad \Delta V = 0.1 + 0.1 = 0.2 \text{ cm}^3.$$

Step 2:

$$\Delta V/V = 0.2/5.0 = 4.0\%, \quad \Delta m/m = 0.1/10.0 = 1.0\%.$$

Step 3:

$$\Delta\rho/\rho \approx 1.0\% + 4.0\% = 5.0\%.$$

Worked derivation:

$$\rho = m/(V_2 - V_1), \quad \epsilon_\rho \approx 0.01 + 0.04 = 0.05.$$

Why: Subtracting similar readings leaves a smaller denominator but does not cancel their worst-case absolute errors. Calculating a percentage against the original 50 or 55 cm3 would substantially underestimate volume uncertainty. The requested first-order convention also differs from an exact nonlinear confidence interval.

Why not the others: (A) combines the correct relative errors in quadrature instead of the requested maximum-error sum; (C) omits mass uncertainty; (D) omits the displacement measurement entirely.

Key point: Propagate subtraction in absolute units before converting to relative error.

Q48 / C Terminal drag calibration followed by an imposed velocity reversal

Answer: (C) g downward.

Given: Buoyancy is unchanged; at the reversed instant speed magnitude is unchanged but drag direction reverses.

Formula/Principle: Use the original terminal balance to determine drag magnitude, then sum forces with the new direction.

Step 1: Let sphere mass be m. Its displaced fluid has mass m/2, so buoyancy is mg/2. The downward effective weight is

$$W_{\text{eff}} = mg - mg/2 = mg/2.$$

Step 2: Originally, drag magnitude at vT equals mg/2 upward. With velocity upward, equal-speed drag is mg/2 downward.

Step 3:

$$F_{\text{net,down}} = mg/2 + mg/2 = mg \Rightarrow a = g.$$

Worked derivation:

$$a_{\text{immediate}} = 2g(1 - \rho_f/\rho_s) = g.$$

Why: Terminal balance holds only for the original downward motion. Reversing velocity reverses drag, not buoyancy or gravity. The result is an instantaneous acceleration, not a claim that it remains g as the speed subsequently changes.

Why not the others: (A) retains the old drag direction; (B) ignores drag after reversal; (D) ignores buoyancy in the terminal balance.

Key point: Re-evaluate force directions after a velocity reversal even when speed is unchanged.

Q49 / D Translation cancellation with nonzero couple torque

Answer: (D) Zero translation; angular acceleration 12F/(ML).

Given: The equal opposite forces are separated by the full rod length L.

Formula/Principle: Centre-of-mass motion uses total force; rotation uses the net moment about the centre.

Step 1:

$$\sum \mathbf{F} = 0 \Rightarrow a_{\text{CM}} = 0.$$

Step 2: Both torques have the same rotational sense:

$$\tau = F(L/2) + F(L/2) = FL.$$

Step 3:

$$\alpha = \tau/I = FL/(ML^2/12) = 12F/(ML).$$

Worked derivation:

$$a_{\text{CM}} = 0, \quad \alpha = 12F/(ML).$$

Why: Opposite forces cancel in translation but their separated lines of action form a couple. Counting just one end's moment halves the actual torque. Zero net force therefore does not imply zero rigid-body motion about its centre.

Why not the others: (A) keeps an uncancelled force and discards the couple; (B) adds force magnitudes rather than vectors; (C) counts only one torque.

Key point: Net force and net torque are independent balances.

Q50 / A Free versus reversible adiabatic expansion to the same volume

Answer: (A) 4.

Given: Neither sample exchanges heat, but their work differs.

Formula/Principle: Use the first law for free expansion and the reversible adiabatic temperature-volume relation only for Q.

Step 1: For P, Qheat=0 and W=0 because external pressure is zero. Thus delta U=0 and ideal-gas TP=T0.

Step 2: For Q,

$$T_Q(8V_0)^{\gamma-1} = T_0V_0^{\gamma-1} \Rightarrow T_Q = T_0/8^{2/3} = T_0/4.$$

Step 3:

$$T_P/T_Q = 4.$$

Worked derivation:

$$T_P = T_0, \quad T_Q = T_0/4.$$

Why: Adiabatic means no heat transfer, not necessarily reversibility. Free expansion has no pressure work on the surroundings, whereas the reversible piston expansion converts internal energy into work. Applying TV^(gamma-1)=constant to both would erase that physical distinction.

Why not the others: (B) assumes equal heat transfer implies equal work; (C) reverses the ratio; (D) uses an exponent one instead of gamma-1.

Key point: Check process reversibility before using an adiabatic state relation.

Q51 / C First-passage time from displacement and velocity sign

Answer: (C) T/6.

Given: Use x=A cos(theta), with theta increasing at angular rate omega=2pi/T.

Formula/Principle: Both displacement and velocity sign select the initial phase.

Step 1:

$$\cos\theta_0 = 1/2, \quad v = -A\omega\sin\theta_0 < 0 \Rightarrow \theta_0 = \pi/3 \pmod{2\pi}.$$

Step 2: The first later phase with cos(theta)=-1/2 is 2pi/3.

Step 3:

$$\Delta t = \frac{2\pi/3 - \pi/3}{2\pi/T} = T/6.$$

Worked derivation:

$$\Delta\theta = \pi/3, \quad \Delta t = T/6.$$

Why: Equal and opposite displacements are not necessarily half a period apart for the first passage. The velocity sign removes the other initial phase, which would give a longer route. SHM spends unequal fractions of time in equal spatial intervals.

Why not the others: (A) uses phase zero instead of the measured initial phase; (B) assumes every sign reversal of displacement needs half a cycle; (D) takes a longer phase route rather than the first arrival.

Key point: Use direction of motion to select the correct SHM phase branch.

Q52 / B Harmonic spacing plus centre-node excitation selection

Answer: (B) 200 Hz.

Given: Consecutive fixed-string harmonics differ by the fundamental frequency.

Formula/Principle: Infer fundamental spacing, then apply the spatial node condition before choosing a harmonic.

Step 1:

$$f_1 = 160 - 120 = 40 \text{ Hz}; \quad 120 = 3f_1, \quad 160 = 4f_1.$$

Step 2: At the centre, harmonic n has displacement factor

$$\sin(n\pi/2),$$

which vanishes for even n.

Step 3: The next odd harmonic above four is five, giving

$$f_5 = 5(40) = 200 \text{ Hz}.$$

Worked derivation:

$$n_{\text{allowed}} = 1, 3, 5, \ldots, \quad f_{\text{next}} = 200 \text{ Hz}.$$

Why: A natural frequency can exist even when a particular driver location cannot excite it. The frequency spacing identifies the complete harmonic series, while the centre condition selects a subset. Merely choosing a large multiple of a supplied frequency misses that spatial information.

Why not the others: (A) is not a harmonic of 40 Hz; (C) is the centre-node sixth harmonic; (D) is the centre-node eighth harmonic.

Key point: Combine resonance frequencies with mode shape at the driving point.

Q53 / C Outer-face flux from an eight-cube symmetry construction

Answer: (C) q/(24 epsilon0).

Given: F is one of four congruent squares making up one outer face of the assembled cube; it does not contain the charge.

Formula/Principle: Apply Gauss's law to the complete symmetric surface, then use symmetry for its face subdivisions.

Step 1: The large cube encloses q strictly inside:

$$\Phi_{\text{large}} = q/\varepsilon_0.$$

Step 2: Centre symmetry gives each of its six faces flux

$$q/(6\varepsilon_0).$$

Step 3: Each outer face divides into four equal-flux congruent squares about its centre, so

$$\Phi_F = \frac{1}{4}\frac{q}{6\varepsilon_0} = \frac{q}{24\varepsilon_0}.$$

Worked derivation:

$$24\Phi_F = q/\varepsilon_0.$$

Why: Gauss's law alone gives total closed-surface flux, not an arbitrary open-face share. The assembly places the charge inside a symmetric closed surface and removes any ambiguity about treating a charge on a small cube's boundary. Symmetry then justifies the factor of twenty-four.

Why not the others: (A) gives a whole large face; (B) gives all three outer faces of one small cube together; (D) divides by an extra factor two without a symmetry reason.

Key point: Embed a boundary-charge geometry in a larger symmetric Gaussian surface.

Q54 / A Thermal bridge parameter inversion and total electrical power

Answer: (A) beta=0.008 per degree C; power 1.32 W.

Given: The equal left divider puts its midpoint at half the supply voltage. At balance no galvanometer current flows.

Formula/Principle: A balanced right divider also has equal arms; then calculate the two series branches in parallel.

Step 1:

$$R_{\text{upper}} = 100(1 + 0.004 \times 50) = 120\ \Omega.$$

Step 2:

$$200(1 - 50\beta) = 120 \Rightarrow \beta = 0.008\ ^\circ\text{C}^{-1}.$$

Step 3: Branch resistances are 200 ohm left and 240 ohm right.

$$I_{\text{total}} = 12/200 + 12/240 = 0.11\ \text{A}.$$

Step 4:

$$P = 12(0.11) = 1.32\ \text{W}.$$

Worked derivation:

$$P = 12^2(1/200 + 1/240) = 0.72 + 0.60 = 1.32\ \text{W}.$$

Why: Balance relates ratios within the bridge, not equality of every arm. The right pair reaches a common 120 ohm while the left stays at 100 ohm. The source supplies both complete branches; the zero-current galvanometer is not another power-dissipating series element.

Why not the others: (B) equates coefficient magnitudes and keeps an incorrect cold-equivalent resistance; (C) doubles the already total source power; (D) divides the resistance change by 100 rather than the lower arm's initial 200.

Key point: Infer thermal parameters from balance, then recompute actual source loading.

Q55 / D Two-time mixture inversion followed by later activity fraction

Answer: (D) 1/3.

Given: Let initial activities be a,b; activity, not initial atom count, is supplied by the detector.

Formula/Principle: Add exponentially decaying activities and solve for their initial mixture before evolving it further.

Step 1:

$$a + b = 900, \quad a/4 + b/2 = 300.$$

Step 2: These give a=600 and b=300.

Step 3: At 4 h,

$$A_A = 600/16 = 37.5, \quad A_B = 300/4 = 75.$$

Step 4:

$$f_A = 37.5/(37.5 + 75) = 1/3.$$

Worked derivation:

$$(A_A : A_B)_{4h} = 1 : 2.$$

Why: Different decay constants change mixture composition over time, so the initial fraction cannot be carried forward unchanged. The second total reading resolves the unknown initial contributions. Equal efficiency does not mean equal source activities or equal numbers of parent nuclei.

Why not the others: (A) gives the initial A activity fraction; (B) assumes equal final components; (C) uses a survival factor rather than a fraction of the mixed final activity.

Key point: Resolve an unknown radioactive mixture before asking for a later component share.

Q56 / B Open-circuit flux change with nonzero emf and zero steady current

Answer: (B) Nonzero induced emf; no sustained conduction current through the gap.

Given: The area and field direction are fixed, and only field magnitude changes.

Formula/Principle: Faraday's law determines induced emf from flux change; a closed conducting route is separately required for sustained conduction.

Step 1:

$$|\mathcal{E}| = \left|\frac{d\Phi_B}{dt}\right| = A|dB/dt| = A|b|.$$

Step 2: Under the specified no-leakage, no-transient approximation,

$$I_{\text{conduction}} = 0.$$

Worked derivation:

$$\mathcal{E} \neq 0 \quad \text{and} \quad I = 0$$

are compatible for an open circuit.

Why: Opening the path removes continuous charge transport but does not remove the changing magnetic flux or the induced electric field. Lenz's law supplies a response direction when current can flow; it does not bridge an ideal insulator. The transient exclusion avoids confusing a brief charge redistribution with a steady current.

Why not the others: (A) identifies absence of current with absence of emf; (C) ignores the insulating interruption; (D) contradicts Faraday's law while inventing an unpowered current.

Key point: Induced emf and closed-circuit current are distinct quantities.

Q57 / C Contact doublet power followed by imaging and orientation

Answer: (C) 60 cm right; magnification -2.

Given: Powers add for thin lenses in contact; use Cartesian signs with initial light to the right.

Formula/Principle: Find equivalent focal length, then apply lens imaging and magnification.

Step 1:

$$1/F = 1/10 + 1/(-20) = 1/20 \Rightarrow F = 20 \text{ cm}.$$

Step 2: With u=-30 cm,

$$1/v - 1/u = 1/F \Rightarrow 1/v = 1/20 - 1/30 = 1/60.$$

Step 3:

$$v = +60 \text{ cm}, \quad m = v/u = 60/(-30) = -2.$$

Worked derivation:

$$F = 20 \text{ cm}, \quad (v, m) = (60 \text{ cm}, -2).$$

Why: The negative lens weakens the converging combination; focal lengths themselves must not be directly added. The object remains outside the equivalent focus, so a real inverted image forms on the opposite side. Magnification sign records orientation, not just enlargement.

Why not the others: (A) mistakes equivalent focal length for image distance; (B) gets location but loses inversion; (D) directly combines focal-length magnitudes into a false symmetric imaging condition.

Key point: Combine powers before applying a signed object-image relation.

Q58 / A Equal-current frequencies with opposite reactance branches

Answer: (A) sqrt2 f.

Given: Equal current and voltage mean equal impedance magnitudes, with the same R at both frequencies. Let omega=2pi f, X=omega L and Y=1/(omega C), both positive.

Formula/Principle: Equal squared reactances allow equal or opposite signed net reactances; check which branch is physically possible.

Step 1:

$$(X-Y)^2=(2X-Y/2)^2.$$

Step 2: The equal-sign branch would give X+Y/2=0, impossible for positive L,C. Thus

$$X-Y=-(2X-Y/2)\Rightarrow Y=2X.$$

Step 3:

$$1/(\omega C)=2\omega L\Rightarrow \omega_0^2=1/(LC)=2\omega^2.$$

Worked derivation:

$$f_0=\sqrt{f(2f)}=\sqrt{2}f.$$

Why: Equal currents on opposite sides of resonance need not have the same phase. The absolute impedance condition hides a sign choice that must be resolved using positivity. Averaging frequencies arithmetically would ignore the inverse-frequency capacitor term.

Why not the others: (B) takes an arithmetic rather than geometric midpoint; (C) assumes the second measured frequency itself is resonance; (D) chooses a frequency below both equal-current points.

Key point: Keep the sign of reactance while using equal impedance magnitudes.

Q59 / D Two-reference diode clipping with a loaded output

Answer: (D) +3 V, -2 V; 2 mA through U.

Given: Conducting ideal diodes have zero forward drop; blocked ones have zero current. The two resistors alone form a half-voltage divider.

Formula/Principle: Propose an on/off state, solve voltages, and verify each diode's bias before applying node-current balance.

Step 1: For +8 V, the off-state divider would give VO=+4 V, forward-biasing U. U clamps O at +3 V, while L has its anode below its cathode and is off.

Step 2:

$$I_{\text{input}}=(8-3)/1000=5\ \text{mA},\quad I_{\text{load}}=3/1000=3\ \text{mA}.$$

Thus $I_U=5-3=2$ mA.

Step 3: For -6 V, the off-state divider gives -3 V, forward-biasing L. It clamps O at -2 V; U is reverse-biased.

Worked derivation:

$$V_O=\text{max}[-2,\text{min}(3,V_{\text{in}}/2)],\qquad I_U(+8)=2\ \text{mA}.$$

Why: The load current is not negligible, so the entire input-resistor current does not flow through the clamping diode. Each reference acts only when its diode would be forward biased; neither diode can be assumed always conducting. Checking the unused diode confirms the proposed states are self-consistent.

Why not the others: (A) ignores both clamps; (B) assigns all input current to U; (C) forgets the lower clamp for negative input.

Key point: Verify diode states and node currents separately in a piecewise circuit.

Q60 / D Restored two-beam interference after common polarisation projection

Answer: (D) 2I0 and zero.

Given: Both beams retain mutual coherence, and their projections onto the common analyser axis overlap in the same detected spatial mode.

Formula/Principle: Apply intensity projection to each beam before adding their same-polarisation amplitudes.

Step 1:

$$I_1' = I_2' = I_0 \cos^2 45^\circ = I_0/2.$$

Step 2: After the analyser,

$$I(\phi) = I_1' + I_2' + 2\sqrt{I_1' I_2'} \cos\phi = I_0(1 + \cos\phi).$$

Step 3:

$$I_{\max} = 2I_0, \quad I_{\min} = 0.$$

Worked derivation:

$$(\sqrt{I_0/2} \pm \sqrt{I_0/2})^2 = 2I_0,\ 0.$$

Why: Orthogonal polarisations give no ordinary interference term before analysis, but a common projection makes the transmitted components parallel. Their amplitudes can then reinforce or cancel because coherence is stipulated. Projection loss and amplitude interference must both be included; the local maximum does not imply an increase of total integrated energy.

Why not the others: (A) adds projected intensities without coherent reinforcement at the maximum; (B) ignores analyser loss; (C) incorrectly retains orthogonality after common projection.

Key point: Check polarisation compatibility before applying coherent intensity addition.